

## Lesson 26 Cardinality of Infinite Sets

Prove that two given sets have the same cardinality by showing that there is a bijection between them.

1.  $\mathbb{R}$  and  $(0, \infty)$
2.  $\mathbb{Z}$  and the set of even integers
3.  $\mathbb{Z}$  and the set of odd integers
4.  $\mathbb{N}$  and the set of even integers
5.  $\mathbb{N} \times \{0, 1\}$  and  $\mathbb{N}$
6. The set of multiples of 3 and  $B$  be the set of multiples of 5
7. The set of positive numbers  $\mathbb{R}_+$  and the set of negative numbers  $\mathbb{R}_-$